

$$\tilde{H}^{\mu\nu} \equiv -\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} H_{\rho\sigma}$$

$$H_{\mu\nu}^{\pm} \equiv \frac{1}{2} (H_{\mu\nu} \pm \tilde{H}_{\mu\nu})$$

Gauge field & Complex scalar:

Here we couple a complex scalar $Z(x)$ to the Gauge field $A_{\mu}(x)$.

$$\mathcal{L} = -\frac{1}{4} \text{Im}(Z) F_{\mu\nu} F^{\mu\nu} - \frac{1}{8} \text{Re}(Z) \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$$

* Symm: $SL(2, \mathbb{R})$ Symmetry

$$\partial_{\mu} \tilde{F}^{\mu\nu} = 0 \quad \& \quad \partial_{\mu} [\text{Im}(Z) F^{\mu\nu} + i \text{Re}(Z) \tilde{F}^{\mu\nu}] = 0$$

We define:

$$G^{\mu\nu} \equiv \epsilon^{\mu\nu\rho\sigma} \frac{\delta \mathcal{L}}{\delta F_{\rho\sigma}} = -i \text{Im}(Z) \tilde{F}^{\mu\nu} + \text{Re}(Z) F^{\mu\nu}$$

Note:

$$G^{\mu\nu-} \equiv Z F^{\mu\nu-} \quad \& \quad G^{\mu\nu+} = \bar{Z} F^{\mu\nu+}$$

$$S = \begin{pmatrix} d & c \\ b & a \end{pmatrix}, \quad ad - bc = 1, \quad S \in SL(2, \mathbb{R})$$

$$\begin{pmatrix} F'^{-} \\ G'^{-} \end{pmatrix} = S \begin{pmatrix} F^{-} \\ G^{-} \end{pmatrix} \quad \Bigg| \quad \begin{pmatrix} F'^{+} \\ G'^{+} \end{pmatrix} = S \begin{pmatrix} F^{+} \\ G^{+} \end{pmatrix}$$

Since $H_{\mu\nu}^{\pm} = (H_{\mu\nu}^{\mp})^*$

$$\begin{pmatrix} d & c \\ b & a \end{pmatrix} \begin{pmatrix} F^{-} \\ G^{-} \end{pmatrix} = \begin{pmatrix} dF^{-} + cG^{-} \\ bF^{-} + aG^{-} \end{pmatrix}$$

Assuming $G'^{\mu\nu-} = Z' F'^{\mu\nu-}$

$$bF^{-} + aG^{-} = Z' (dF^{-} + cG^{-})$$

But $G^{-} = Z F^{-}$

$$bF^{-} + aZ F^{-} = Z' (dF^{-} + cZ F^{-})$$

$$\therefore Z' = (b + aZ) / (d + cZ)$$

Coupled Maxwell fields: $A_{\mu}^A(x)$ $A=1, \dots, 2, m$

$$f_{AB}(\phi) = f_{AB}(\phi) \in \mathbb{C}$$

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$$S = \int d^4x \mathcal{L}, \quad \mathcal{L} = -\frac{1}{4} \text{Re}(f_{AB}) F_{\mu\nu}^A F_B^{\mu\nu} + \frac{1}{4} i \text{Im}(f_{AB}) F_{\mu\nu}^A \tilde{F}^{\mu\nu B}$$

Here we have:

$$\begin{pmatrix} F'^{-} \\ G'^{-} \end{pmatrix} = S \begin{pmatrix} F^{-} \\ G^{-} \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} F^{-} \\ G^{-} \end{pmatrix}, \quad A \in M_{m \times m}(\mathbb{R})$$

Note: $G_A^{\mu\nu-} = i f_{AB} F^{\mu\nu-B}$

$$G'^{-} = C F^{-} + D G^{-} = (C + i D f) F^{-} \quad \leftarrow$$

$$F'^{-} = A F^{-} + i f B F^{-} = (A + i f B) F^{-}$$

$$(A + i f B)^{-1} F'^{-} = F^{-}$$

$$G'^{-} = (C + i D f) (A + i f B)^{-1} F'^{-}$$

$$\therefore i f' = (C + i D f) (A + i f B)^{-1}$$

Note that $i f'$ is symmetric, by uac.

$$(i f') = (i f')^T$$

$$(i f')^T = (A + i f B)^{T-1} (C^T + i f^T D^T)$$

Mult. By $(A + i f B)$ on the Right:

$$(C + i D f) = (A + i f B)^{T-1} (C^T + i f^T D^T) (A + i f B)$$

Mult. the Left By $(A + i f B)^T$

$$(A^T + i f^T D^T) (C + i D f) = (C^T + i f^T D^T) (A + i f B)$$

$$A^T C + A^T i D f + i f^T B^T C - f^T B^T D f$$

$$= C^T A + i C^T f B + i f^T D^T A - f^T D^T B f$$

$$A^T C - C^T A = 0$$

$$B^T D - D^T B = 0$$

$$A^T D f + f^T B^T C = C^T B f + A^T D A$$

$$(A^T D - C^T B) f = f (D^T A - B^T C)$$

$$\therefore A^T D - C^T B = 0$$

These are the defining conditions of a symplectic group element in $\dim = 2m$:

$$S = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{Sp}(2m, \mathbb{R})$$

$$S^T \Omega S = \Omega, \quad \Omega = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$$